

London School of Economics
EC400 Mathematics Examination
September 2012

Not for MSc in Finance and Economics Students
Time: 3 hours

This examination has three parts: Revision Mathematics, Mathematics for Microeconomics, and Mathematics for Macroeconomics. Each part is worth 100 marks. Write your name, your class group number, and the name of your class teacher on your script. Explain your answers to get high marks.

Part 1. Revision Mathematics. Answer all questions.

Some questions have yes/no answers, or numerical answers. Explain your answers to these questions.

1. (a) ...

Part 2. Mathematics for Microeconomics. Answer all questions.

5. (20 marks) Consider the following quadratic form:

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2cx_1x_3$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the leading principal submatrices of A .

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *negative semi-definite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is negative semi-definite?
- (c) (8 marks) For what values of the parameters a, b, c , is the quadratic form $Q(\mathbf{x})$ *positive definite*? For what values is it *negative definite*? For what values is it *positive semi-definite*?

6. (40 marks) For some constant $k > 0$ consider the function defined on $\mathbb{R}^2$:

$$f(x, y) = 2ky(x + 1) - k^2(x^4 + 2x^2)/2 - y^2.$$

- (a) (8 marks) Find the *critical points* of $f(x, y)$.
- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?
- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?
- (d) (8 marks) Is $f(x, y)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above*?
- (e) (8 marks) Denote the *global maximizer* of $f(x, y)$ by $(x(k), y(k))$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in k . Also show how the *global maximizer* $(x(k), y(k))$ is affected by marginal changes in k .

7. (40 marks) For $p > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & F(x, y) = x^{3/4}y^{1/4} \\ \text{s.t.} \quad & x + py \leq 10 \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \tag{1}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?
- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can we solve this optimization problem using the Kuhn-Tucker method? Are the Kuhn-Tucker constraint qualifications satisfied? Is it possible to solve the same optimization problem with the Lagrangian method? Are the Lagrange constraint qualifications satisfied?
- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions for the solution of the constrained optimization problem.
- (d) (4 marks) Are these conditions also *sufficient* to characterize an optimum in this problem?
- (e) (4 marks) Solve the constrained optimization problem in terms of p .
- (f) (8 marks) Let $x(p)$ and $y(p)$ be the solution of this problem. Find the slope of such functions with respect to p . Moreover, find how the maximum value function is affected by a change in p .

Part 3. Mathematics for Macroeconomics. Answer all questions.

8. (50 marks) Let $\alpha \in (0, 1)$, $\delta \in (0, 1)$, $\rho \in (0, 1)$, and $\sigma > 0$. Impose that $\rho + \delta < 1$. Given $k(0)$, we want to find the function $c(t)$ to maximize

$$\int_0^{+\infty} e^{-\rho \cdot t} \cdot \frac{c(t)^{1-\sigma} - 1}{1 - \sigma} dt,$$

subject to the law of motion

$$\dot{k}(t) = k(t)^\alpha - c(t) - \delta \cdot k(t).$$

- (a) (20 marks) Which variable do you choose as a state variable? Which variable do you choose as a control variable? Write down the current-value Hamiltonian and derive the optimality conditions.
- (b) (5 marks) The Euler equation is the first-order differential equation that characterizes the optimal function $c(t)$. Determine the Euler equation.
- (c) (10 marks) Suppose $\alpha = 1$ and $\sigma = 1$. Show that the system describing the optimal functions $\{k(t), c(t)\}$ reduces to a linear, homogenous system of first-order differential equations. Show that the system is unstable by computing the eigenvalues.
- (d) (15 marks) Suppose $\alpha \in (0, 1)$ and $\sigma > 0$. Show that the system describing the optimal functions $\{k(t), c(t)\}$ reduces to a nonlinear system of first-order differential equations. Use a phase-diagram to show that the steady state of the system is a saddle point. Explain how you draw the phase diagram.

9. (50 marks) Let $\beta \in (0, 1)$ and $r > 0$. Given $k_0 > 0$, we want to find a collection of sequences $\{c_t, k_{t+1}\}_{t=0}^{+\infty}$ to maximize

$$\sum_{t=0}^{\infty} \beta^t \cdot \ln(c_t),$$

subject to the constraints

$$k_{t+1} = (1 + r) \cdot k_t - c_t$$

for all $t \geq 0$.

- (a) We first solve the maximization problem using the Lagrangian method.
 1. (5 marks) Write down the Lagrangian of the problem.
 2. (5 marks) Derive the first-order condition(s) of the maximization problem.
 3. (5 marks) Derive the Euler equation.
- (b) We now solve the maximization problem using the dynamic programming method.
 1. (5 marks) Which variable do you choose as a state variable? Which variable do you choose as a control variable? Write down the Bellman equation.
 2. (5 marks) Derive the first-order condition associated with the Bellman equation.
 3. (5 marks) Derive the Benveniste-Scheinkman equation.
 4. (5 marks) Derive the Euler equation. Compare it with the Euler equation obtained with the Lagrangian method and discuss.
 5. (5 marks) Suppose that the policy function for consumption takes the form $h(k) = A \cdot (1 + r) \cdot k$ where $A \in (0, 1)$. Derive A .
 6. (10 marks) Suppose that the value function takes the form $V(k) = B + D \cdot \ln(k)$, where B and D are constants. Using the expression for the policy function that you derived in the previous question, derive B and D .